

JEE MAIN REVISION MASTER: DEFINITE INTEGRATION SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 322 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Definite Integration - Core Application	322 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Definite Integration - Core Application (Very High Yield — 322 PYQs)

Core Principle: Key problem pattern in Definite Integration.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The value of the integral $\int_0^2 \frac{\sqrt{x(x^2+x+1)}(\sqrt{x+1})(\sqrt{x^4+x^2+1})}{dx}$ is equal to:..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "The value of the integral $\int_{-(\pi/4)}^{(\pi/4)} (\frac{32\cos^4 x + e^{\sin x}}{dx}) dx$ is :..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "If $\alpha=1$ and $\beta=1+i\sqrt{2}$, where $i=\sqrt{-1}$ are two roots of the equation $x^3+ax^2+bx+c=0, a, b, c \in \mathbb{R}$, then $\int_{-1}^{\beta} \dots$ " [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

The value of the integral $\int_{-\pi/4}^{\pi/4} (32\cos^4 x + e^x \sin x) dx$ is :

- (A) $4\pi+2$
- (B) $3\pi+8$ **[Correct]**
- (C) $3\pi+4$
- (D) $4\pi+3$

Correct Answer: B

Solution Analysis: Let $I = \int_{-\pi/4}^{\pi/4} \frac{32 \cos^4 x + e^{\sin x}}{1 + e^{\sin x}} dx$. We will use the standard property: If $I = \int_{-a}^a f(x) dx$, then $I = \int_{-a}^a f(-x) dx$ and hence $2I = \int_{-a}^a (f(x) + f(-x)) dx$. Now here, $f(x) = \frac{32 \cos^4 x + e^{\sin x}}{1 + e^{\sin x}}$. So, $f(-x) = \frac{32 \cos^4(-x) + e^{\sin(-x)}}{1 + e^{\sin(-x)}}$. Using $\cos(-x) = \cos x$ and $\sin(-x) = -\sin x$, $f(-x) = \frac{32 \cos^4 x + e^{-\sin x}}{1 + e^{-\sin x}}$. Therefore, $f(x) + f(-x) = \frac{32 \cos^4 x (1 + e^{\sin x}) + (1 + e^{\sin x})(1 + e^{-\sin x})}{(1 + e^{\sin x})(1 + e^{-\sin x})}$. Now simplify the bracket: $\frac{32 \cos^4 x + 1 + e^{\sin x} + e^{-\sin x}}{1 + e^{\sin x} + e^{-\sin x} + 1}$. Let $t = e^{\sin x}$. Then $e^{-\sin x} = (1/t)$. So, $\frac{32 \cos^4 x + 1 + t + 1/t}{1 + t + 1/t + 1}$.